DP Electronics (Deutsche Power)

DP Electronics (Deutsche Power)



Lead Acid Replacement

12V/24V LiFePO4 Battery





Features

- lead-acid plastic shell package
- Higher energy density.
- Weight lighter, only 30% of Lead acid Battery
- Built-in battery charge/discharge protection
  Intelligent battery balance management with BMS
- Longer Life 3000 – 5000 Cycles(100%DOD)

Application

- Solar/Wind energy storage system
  Telecommunication
- Residential/Hospital/School/Factory standby power
- Electric power systems

Lead Acid Replacement Battery-Parameters

Item	Parameter			
Model	CPLI-12.8V100Ah	CPLI-12.8V200Ah	CPLI-25.6V100Ah	CPLI-25.6V200Ah
Battery Type	Lithium Iron Phosphate			
Energy	1.28KWh	2.56KWh	2.56KWh	5.12KWh
Nominal Voltage	12.8V	12.8V	25.6V	25.6V
Nominal Capacity	100Ah	200Ah	100Ah	200Ah
Factory Voltage	11V~12.8V		22~25.6V	
Charge Method	CC-CV 14.6V 0.05C cut off		CC-CV 29.2V 0.05C cut off	
Max. Continuous Charge Current	100A			
End of Discharge Voltage	10V		20V	
Max Continuous Discharge Current	100A			
Cycle Life	≥5000times 80%SOC 0.5C/0.5C 25℃			
ACIR	≤12mΩ			
Max. Series/Parallel	4P/No Support Series			
Charge Temperature	0℃~45℃			
Discharge Temperature	-10℃~45℃			
Storage Temperature	10℃~35℃			
Weight	10KG	19.5KG	19.5KG	38KG
Size	L327*W172*H242mm	L523*W240*H224mm	L523*W240*H224mm	L523*W240*H224mm
Certification	IEC62619 UN38.3 CE RoHS			

Rackmount | Powerwall | Vertical

Rackmount Module
48V/51.2V LiFePo4 Battery







Features

- **Long Cycle Life:** 3000~5000 times,service more than 8 to10 years.
- **Installation:** The battery size shall conform to 19 inch standard cabinet.
- **High Power:** Support multiple batteries in series or in parallel to achieve greater capacity and charging and discharging current; Up to 16 in parallel are supported
- **Safety:** Level 4 protection, cell protection, BMS protection, air switch protection, grounding protection.
- **Communication:** The battery supports the communication protocol of 90% inverter manufacturers on the market.

BMS
Buildin



Application

- Solar/Wind energy storage system Telecommunication
- Residential/Hospital/School/Factory standby power
- Electric power systems

Model	CLFP-51.2-50-R	CLFP-51.2-100-R	CLFP-51.2-200-R
Battery Type	Lithium Iron Phosphate		
Combination method	3.2V52Ah Cell-16S1P	3.2V102Ah Cell-16S1P	3.2V105Ah Cell-16S2P
Energy	2.56KWh	5.12KWh	10.24KWh
Norminal Voltage	51.2V	51.2V	51.2V
Norminal Capacity	50Ah	100Ah	200Ah
Factory Voltage	52.0V-54.40V		
Charge Method	CC-CV 58.4V 0.05C cut off		
Max. Continuous Charge Current	50A T≥10℃	100A T≥10℃	100A T≥10℃
End of Discharge Voltage	40V		
Max. Continuous Discharge Current	50A T≥10℃	100A T≥10℃	100A T≥10℃
Cycle life	≥3000times 80%SOC 0.5C/0.5C 25℃	≥5000times 80%SOC 0.5C/0.5C 25℃	≥4000times 80%SOC 0.5C/0.5C 25℃
ACIR	≤20mΩ		
Batteries Parallel/Series	Max.16P/No support series		
Charge Temperature Range	0~45℃		
Discharge Temperature Range	-10~55℃		
Storage Temperature Range	Less than 8 months: -10~35℃ 60±25%R.H. less than 3 months: -10~45℃ 60±25%R.H. Less than 7 days: -20~65℃ 60±25%R.H.		
Installation Method	Standard 19inch cabinet		
Communication	RS485 RS232 CAN		
Waterproof	IP21		
Weight	27.5kg	47kg	87kg
Size(W*D*H)	483*370*177mm	483*550*177mm	483*600*266mm
Certification	IEC62619 UN38.3 CE RoHS		

Powerwall Series

5KW/10KW LiFePO4 Battery





Features

- **Long Cycle Life:** 5000 times,service more than 10 years.
- **High Power:** Support long-term charging and discharging of 100A
- **Physical characteristics:** small size, light weight, thin thickness
- **Installation mode:** wall mounted, hanging on the wall together with inverter and control to form a harmonious environment; The left and right sides of the battery and the panel are equipped with handles for easy installation and handling.
- **Safety:** Level 4 protection, cell protection, BMS protection,inverter protection, air switch protection, grounding protection.
- **Communication:** The battery supports the communication protocol of 90% inverter manufacturers on the market.

Application

- Solar/Wind energy storage system
- Community/Villa/Residential/Hospital/School/Gymnasium etc
- House Building/Home use

Powerwall-Parameters

Model	CLFP-51.2-100-M	CLFP-51.2-200-M
Battery Type	Lithium Iron Phosphate	
Combination method	3.2V100Ah Cell-16S1P	3.2V100Ah Cell-16S2P
Energy	5.12KWh	10.24KWh
Norminal Voltage	51.2V	
Norminal Capacity	100Ah	200Ah
Factory Voltage	52.0V-54.40V	
End of Discharge Voltage	58.4V	
Max. Continuous Charge Current	100A T≥10℃	
Charge Voltage	40V	
Max. Continuous Discharge Current	100A T≥10℃	
Cycle life	≥5000times 80%SOC 0.5C/0.5C 25℃	
ACIR	≤10mΩ	≤20mΩ
Batteries Parallel/Series	Max.16P/No Support Series	
Charge Temperature Range	0~45℃	
Discharge Temperature Range	-10~55℃	
Storage Temperature Range	Less than 8 months: -10~35℃ 60±25%R.H. less than 3 months: -10~45℃ 60±25%R.H. Less than 7 day: -20~65℃ 60±25%R.H.	
Installation Method	Wall Mounted	
Communication	RS485 RS232 CAN	
Waterproof	IP54	
Weight	48±2Kg	88±2Kg
Size	W420*D160*H630mm	W510*D160*H855mm
Certification	IEC62619 UN38.3 CE RoHS	

Vertical System

LiFePO4 Battery + Hybrid Inverter



Vertical Battery System

Battery Module		
Model	CLFP-51.2-100-V	CLFP-51.2-200-V
Battery Type	Lithium Iron Phosphate	
Energy	5.12KWH	10.24KWH
Nominal Voltage	51.2V	
Nominal Capacity	100AH	200AH
Charge Method	CC-CV 58.4V 0.05C cut off	
Max.Continuous Charge Current	100A	
End of Discharge Voltage	40V	
Max.Continuous Disharge Current	100A	
Cycle Life	≥5000times 80%SOC 0.5C/0.5C 25℃	
Use Period	> 10 Years@25℃	
Cooling Method	Natural Cooling	
Storage	10℃ ~ 35℃ ≤85%RH	
Operating Temperature	20℃~55℃	
Installation Method	Stacked	
Communication	RS485 RS232 CAN	
Waterproof	IP42	
Size/Weight	L538*W200*H480mm/54KG(single layer)	
Certification	IEC62619 UN38.3 CE RoHS	

Hybrid Inverter	
Phase Number	Single-Phase Sin Pure Wave
Output Power	5KW (Optional)
Rated Grid Frequency	50Hz/60Hz Auto Detect
AC Input/Output Voltage Range	220V
Max.PV Input Power	5000W (Optional)
MPPT Voltage Range	60-115Vdc
Battery Voltage Range	40.0Vdc~60Vdc ± 0.6Vdc
Cooling Method	Fans
Waterproof	IP21
Size/Weight	L538*W200*H390mm/18KG
Certification	IEC62619 UN38.3 CE RoHS

Chassis	
Chassis Size/Weight	L538*W60*H188mm/10KG

Stacked System

$LiFePO_4$ Battery + Hybrid Inverter





Features

- **Long Cycle Life:** 5000 times,service more than 10 years.
- **Modular design:** the battery module can be increased according to the power consumption; The battery system power range is 5kwh~25kwh.
- **Movable:** The battery chassis is equipped with four wheels capable of bearing 500kg weight, which is convenient to push the battery to the parking area on the flat ground. Wheel rotation and locking can be set manually.
- **Integrated design:** 3~5kw off grid integrated machine inverter can be installed on the top of the battery pack as required.
- **Safety:** Level 4 protection, cell protection, BMS protection,inverter protection, air switch protection, grounding protection.

Application

- Solar/Wind energy storage system
- Community/Villa/Residential/Hospital/School/Gymnasium etc

Stacked Battery System-Battery Parameters

Battery Module				
Model	CLFP-51.2-100-S	CLFP-51.2-200-S	CLFP-51.2-300-S	CLFP-51.2-400-S
Battery Type	Lithium Iron Phosphate			
Energy	5.12KWH	10.24KWH	15.36KWH	20.48KWH
Nominal Voltage	51.2V			
Nominal Capacity	100AH	200AH	300AH	400AH
Charge Method	CC-CV 58.4V 0.05C cut off			
Max.Continuous Charge Current	100A			
End of Discharge Voltage	40V			
Max.Continuous Disharge Current	100A			
Cycle Life	⩾5000times 80%SOC 0.5C/0.5C 25℃			
Use Period	> 10 Years@25℃			
Number Of Expandable Batteries	1~15	2~15	3~15	4~15
Optional Capacity Range	5.12~76.8KWH	10.24~76.8KWH	15.36~76.8KWH	20.48~76.8KWH
Cooling Method	Natural Cooling			
Storage	10℃~35℃ ⩽85%RH			
Operating Temperature	20℃~55℃			
Installation Method	Stacked			
Communication	RS485 RS232 CAN			
Waterproof	IP42			
Size/Weight	550*500*170mm/54.5KG(single layer)			
Certification	IEC62619 UN38.3 CE RoHS			
Hybrid Inverter				
Phase Number	Single-Phase Sin Pure Wave			
Output Power	5KW (Optional)			
Rated Grid Frequency	50HZ/60HZ Auto Detect			
AC Input/Output Voltage Range	220V			
Max.PV Input Power	5000W (Optional)			
MPPT Voltage Range	60-115Vdc			
Battery Voltage Range	40.0Vdc~60Vdc			
Cooling Method	Fans			
Waterproof	IP21			
Size/Weight	550*500*170mm/18KG			
Certification	IEC62619 UN38.3 CE RoHS			
Chassis				
Chassis Size/Weight	L550*W500*H40mm/10KG			
Wheels Size/Load-bearing	H92mm/125KG*4			

Stacked Battery System



5 KWh



10 KWh

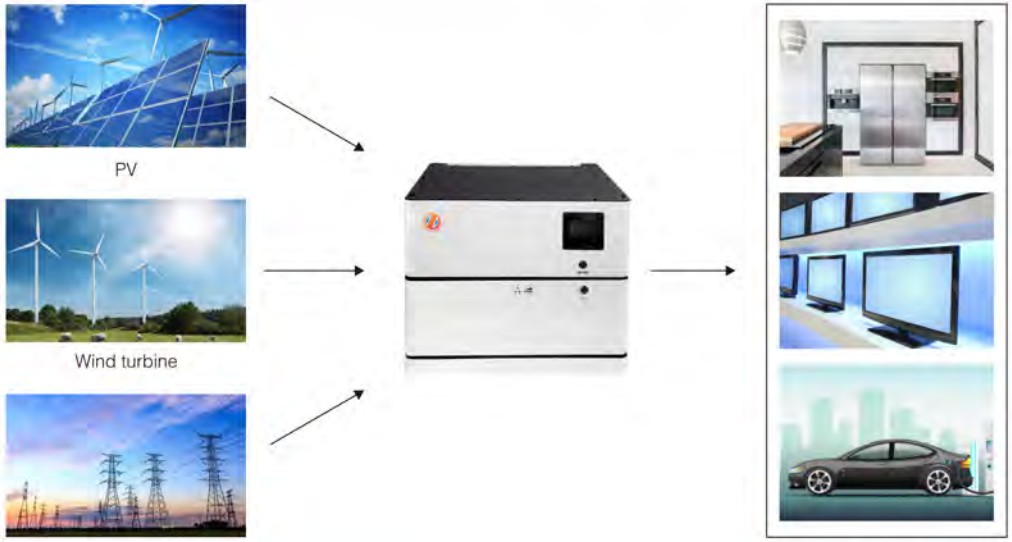


15 KWh

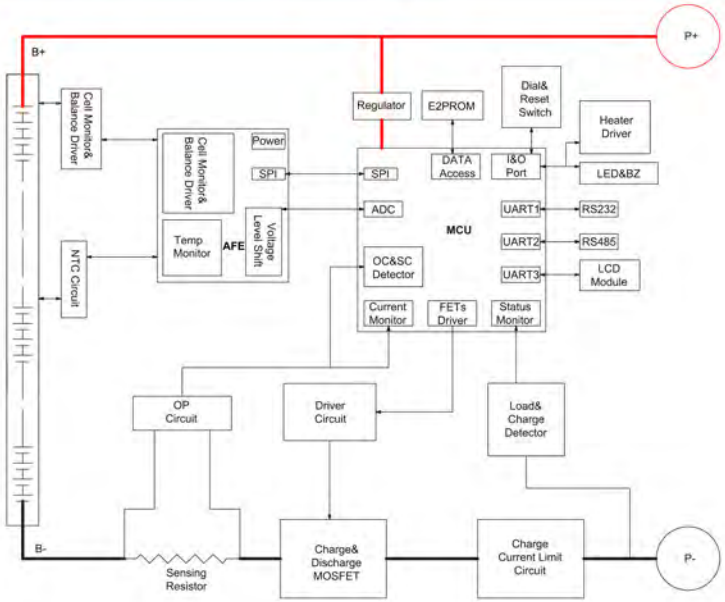


20 KWh

Stacked Battery System–Diagram



Single Pack Function Image



High Voltage System

LiFePO4 Battery +Control Module



Features

- High power
- Lighter weight
- Long cycle life
- Superior safety

Application

- UPS
- Residential
- Hospital
- School
- Factory
- Standby power

HV Battery System-96V/192V

Model	CLFP96-100-H	CLFP96-200-H	CLFP192-100-H	CLFP192-200-H
Nominal voltage	96V		192V	
Nominal capacity	100Ah	200Ah	100Ah	200Ah
Energy	9.6kwh	19.2kwh	19.2kwh	38.4kwh
Cycle life	3500 times@25°C 1C/1C 80%DOD			
Charge voltage	106V		213V	
Standard Charging Current	50A	100A	50A	100A
Max.charge current	100A	200A	100A	200A
Discharge cut-off voltage	75V		150V	
Discharge current	50A	100A	50A	100A
Max.Discharge current	100A	200A	100A	200A
Discharge temperature	-20~55°C			
Charge temperature	0~55°C			
Storage temperature	-10~45°C			
Communication	RS485/CAN			
Display	LCD			
Terminal	Anderson/Connector			
Weight	100kg	180kg	180kg	355kg
Dimension(440*650*170mm)	6U	12U	12U	24U
Warranty	3 years			
Designed life	10 years			
Waterproof	IP54			
Certification	CE UL RoHS UN38.3			

HV Battery System



HV Battery System-480V

Model	CLFP480-40-H	CLFP480-80-H	CLFP480-120-H	CLFP480-40*N-H
Nominal voltage	480V			
Cluster	1	2	3	N
Adjustable rate voltage	384V/480V/499.2V/528V			
Nominal capacity	40Ah	80Ah	120Ah	40Ah*N
Energy	19.2kwh	38.4kwh	57.6kwh	19.2kwh*N
Cycle life	3500 times@25°C 1C/1C 80%DOD			
Charge voltage	532V			
Adjustable charge voltage	426V	532V	553V	585V
Standard Charging Current	40A	80A	120A	40A*N
Max.charge current	80A	160A	240A	80A*N
Discharge cut-off voltage	375V			
Adjustable discharge cut-off voltage	300V/375V/390V/412V			
Discharge current	80A	160A	240A	80A*N
Max.Discharge current	160A	320A	480A	160A*N
Discharge temperature	-20~55°C			
Charge temperature	0~55°C			
Storage temperature	-10~45°C			
Communication	RS232/RS485/CAN			
Display	LCD			
Terminal	Anderson/Connector			
Weight	280kg	560kg	840kg	280kg*N
Dimension(440*630*132mm)	21U*1P	21U*2P	21U*3P	21U*NP
Warranty	3 years			
Designed life	10 years			
Waterproof	IP54			
Certification	CE UL RoHS UN38.3			

HV Battery System



HV Battery System-500V~800V

Model	CLFP537-100-H	CLFP537-200-H	CLFP537-300-H	CLFP537-100*N-H
Nominal voltage	537.6V			
Cluster	1	2	3	N
Adjustable rate voltage	500~800V			
Nominal capacity	100Ah	200Ah	300Ah	100Ah*N
Energy	53.76kwh	107.52kwh	161.28kwh	53.76kwh*N
Cycle life	3000 times@25°C 1C/1C 80%DOD			
Charge voltage	596V			
Adjustable charge voltage	500V~900V			
Standard Charging Current	50A	100A	150A	50A*N
Max.charge current	100A	200A	300A	100A*N
Discharge cut-off voltage	420V			
Adjustable discharge cut-off voltage	400V~800V			
Discharge current	50A	100A	150A	50A*N
Max.Discharge current	100A	200A	300A	100A*N
Discharge temperature	-20~55°C			
Charge temperature	0~55°C			
Storage temperature	-10~45°C			
Communication	RS485/CAN			
Display	LCD			
Terminal	Anderson/Connector			
Weight	490kg	980kg	1470kg	490kg*N
Dimension(440*630*132mm)	32U*1P	32U*2P	32U*3P	32U*NP
Warranty	3 years			
Designed life	10 years			
Waterproof	IP54			
Certification	CE UL RoHS UN38.3			

Power Battery

24V-72V $LiFePO_4$ Battery



Suitable for various electric vehicle





Features

- High power
- Lighter weight
- Long cycle life
- Superior safety

Application

- Golf cart
- Trike
- Forklift
- RV

Power Battery-Parameters

Model	CLFP48-100-P	CLFP51.2-100-P
Nominal voltage	48V	51.2V
Nominal capacity	100Ah	100Ah
Energy	4800Wh	5120Wh
Cycle life	3000 times@25°C 1C/1C 80%DOD	
DCIR	< 50mΩ	
Charge voltage	54.75V	58.4V
Standard Charging Current	50A	50A
Max.charge current	100A	100A
Discharge cut-off voltage	37.5V	40V
Discharge current	100A	100A
Max.Discharge current	200A	200A
Max.pluse discharge current(3S)	600A	600A
Discharge temperature	-20~55°C	
Charge temperature	0~55°C	
Storage temperature	-10~45°C	
Display	LCD	
Terminal	M8/Anderson	
Weight	45kg	47kg
Dimension	520*260*230mm	520*260*230mm
Waterproof	IP54	
Warranty	3 years	
Designed life	10 years	
Certification	CE UL RoHS UN38.3	

60V Series(60.8Vdc)	CLFP60 -20-P	CLFP60 -30-P	CLFP60 -40-P	CLFP60 -60-P	CLFP60 -70-P	CLFP60 -80-P	CLFP60 -100-P	CLFP60 -120-P
72V Series(60.8Vdc)	CLFP72 -30-P	CLFP72 -40-P	CLFP72 -60-P	CLFP72 -70-P	CLFP72 -75-P	CLFP72 -80-P	CLFP72 -100-P	CLFP72 -125-P	CLFP72 -140-P	CLFP72 -180-P

Container Energy Storage System

Container Energy Storage System

The energy storage battery system was installed into 20ft or 40ft standard/customized container, containing LiFePO4 battery modules, Battery management system, EMS, PCS, lighting, fire fighting system, in-place monitoring, Cooling system etc.
This CESS can cover different voltage and capacity (300kwh ~ 3MWh) to achieve centralized energy storage. Working with photovoltaic, wind power and auxiliary power to achieve energy consumption, smooth power output, peak shaving and valley filling and provide auxiliary services for the grid.

Case

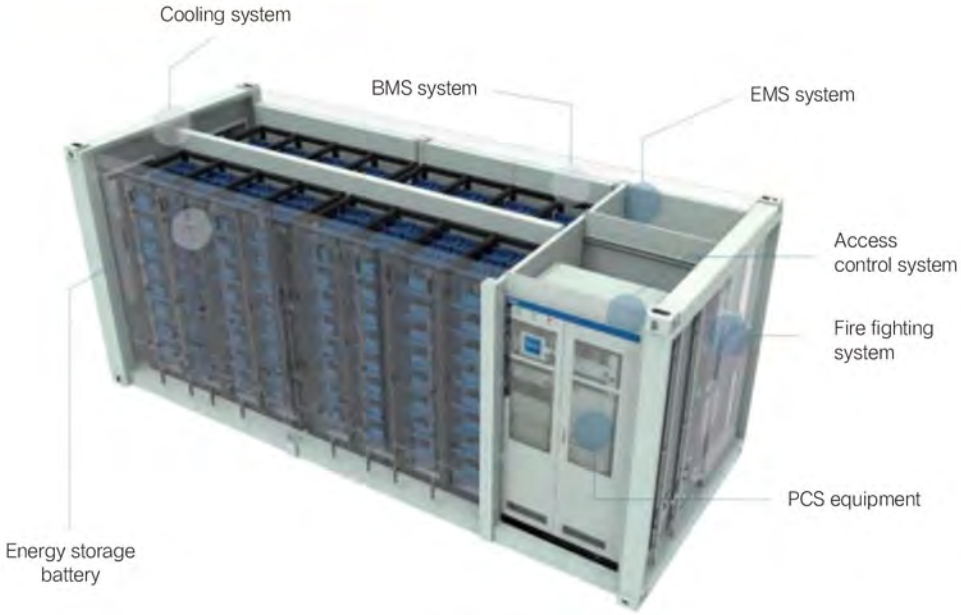
- **High energy:** 2.45MWh
- **High-power:** 1.25MW output power;
  400Vac grid connection;
- **Good heat dissipation:**
  4 sets of 3kw air conditioners



Application

- Industrial area, Factory, Large farms, Islands, Deserts, Docks
- all aspects of generation, transmission, distribution.

CESS Perspective View



CESS – On Grid





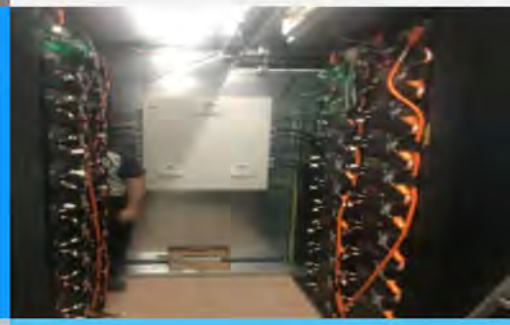
No.	Item	Parameter	Remark
Battery system			
1	System capacity (kwh)	2451.456kwh	
2	Rated DC voltage (V)	729.6	
3	DC system voltage range (V)	615.6~820.8	
AC output parameters			
4	AC access mode	Three phase three wire (without isolating transformer)	
5	Rated power (kw)	2*630	
6	Maximum capacity (kVA)	2*693	
7	Rated grid voltage (V)	400	
8	Rated current (A)	2*910	
9	Maximum operating current (A)	2*1000	
10	Rated grid frequency (Hz)	50	
11	THDi	<3% (Rated power)	
12	Power factor	0.99 (leading) ~0.99 (lagging)	
13	Overload capacity	1.1x long-term	
Others parameters			
14	Communication mode	Ethernet、CAN2.0、RS485	
15	Degree of protection	IP54	
16	Operating temperature (° C)	0~45	
17	Highest altitude (m)	≤5000	
18	Fire fighting system	Heptafluoropropane gas fire extinguishing system	
19	Container size L*W*H	6058*2438*3000mm	

Cabinet/Container ESS Case



Time: 2019
Location: USA
Project composition: 250KW/500KWh
Effect:
- Peak cutting and valley filling
- emergency power supply

Time: 2020
Place: Poland
Project composition: 850kWh
Effect:
- Microgrid



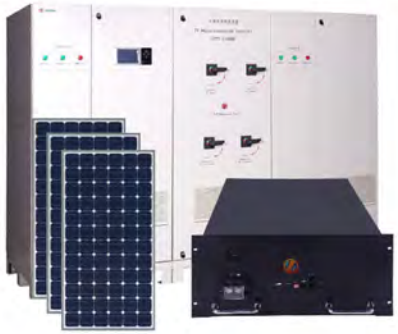


Time: 2022
Location: USA
Project composition: 6.39MWh
Application:
- Energy transfer
- New energy smoothing
- Optical storage DC coupling

Time: 2022
Place: Poland
Project composition: 15.5MWh
Effect:
- Peak cutting and valley filling
- Frequency modulation



Solar Energy Storage System



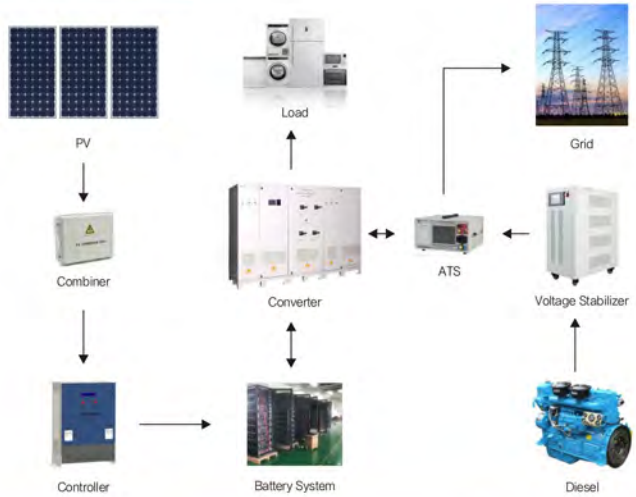
Product Parameters

- **High energy:** 665.6KWh;
- **Large power:** 100KW output ;
- **High voltage output:** support 380Vac high voltage power grid connection; It also contains the output port of 220Vac civil power.
- **System integrity:** the system includes all components from the power generation end (solar panel) to the grid-connected output end.
- **Customization:** According to the user's electricity demand, the appropriate system can be customized.

Application

- Industrial area, Factory
- Schools, Hospitals, Sports centers, Commercial buildings, Communities
- Farms, Islands, Deserts

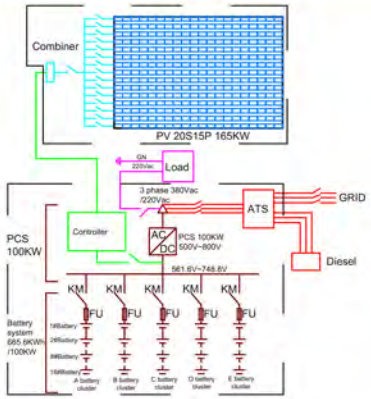
Diagram & Specification



Item	Model	QTY
Battery system	665.6V 1000Ah	1set
Converter	380V/220V 100KW	1pcs
ATS	300Aac	1pcs
Solar panel	550W	300pcs
Solar panel rack	High 1m	300pcs
Solar controller	165KW	1pcs
Combiner box	15 in 1 out	1pcs
Voltage regulator	380Vac 100KW	1pcs

※Solar Energy Battery System

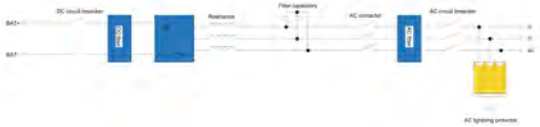




Component	Item	Parameter
Battery cluster	Nominal voltage	665.6V
	Nominal capacity	200Ah
	Nominal energy	133.12kwh
	Module&Method	16S1P
	Weight	1300kg
Battery system	Nominal voltage	665.6V
	Nominal capacity	1000Ah
	Nominal energy	665.6kwh
	Charge voltage	748.8V
	Standard Charging Current	250A
	Discharge end of voltage	561.6V
	Discharge current	250A
	Cluster&Method	5P
	Weight	6500kg

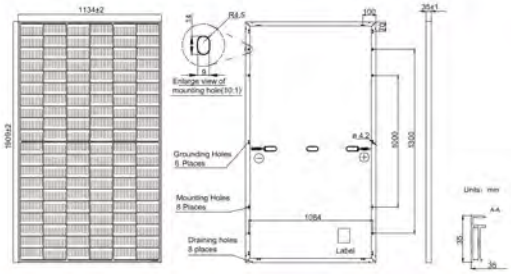
※Converter for Solar Energy System





※Solar panel & Controller





Name	Parameter	Remark
Max Power(W)	550	STC
Open Circuit Voltage(V)	49.95	
Short Circuit Current(A)	14.05	
Voltage at Pmax(V)	41.97	
Current at Pmax(A)	13.11	
Module Efficiency(%)	21.28	
Max.System Voltage	1500Vdc	
Max.series fuse rating	20A	
Operating temperature range	-40~+85°C	
Max.static load front	5400Pa(e.g.snow)	
Max.static load back	2400Pa(e.g.wind)	
Max.hailstone impact	25mm/23m/s(diameter/velocity)	
Weight	27.3kg	
Dimensions	2279*1134*35mm	
Cell	144/Mono	
Front cover	Low-iron tempered glass/3.2mm	
Cable	300mm/4mm²	
Frame	Anodized aluminum alloy	
Junction box	IP68	
Connector	MC4	
Application level	Class A	

Model	MPPT165
Maximum PV open circuit voltage	1000Vdc
PV operating voltage range	500~1000Vdc
MPPT voltage range	550~850Vdc
Battery voltage range	500~800Vdc
Maximum output power	180KW
PV charging current range (can be set)	0~300A